

Estimating and filtering out ambient mRNA for sc-/sn-RNA-Seq Analysis in 10X Genomics data

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```
## The following object is masked _by_ .GlobalEnv:
##
##     root_dir
```

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DNB Bioinformatics Core Pipeline: Standard sc-/sn-RNA-Seq Analysis in 10X Genomics data

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1 Information about this notebook

SoupX method[2] profiles "the soup", i.e., collection of cell-free mRNAs floating in the input solution. The soup looks different for each input solution and strongly resembles the expression pattern obtained by summing all the individual cells (Young and Behjati, 2020[3]).

SoupX calculates `Cell-specific contamination fraction` (Estimate (or manually set) the contamination fraction, the fraction of UMIs originating from the background, in each cell) and infers a `corrected expression matrix` (Correct the expression of each cell using the ambient mRNA expression profile and estimated contamination).

The method to do this consists of three parts:

1. Calculate the profile of the soup.
2. Estimate the cell specific contamination fraction.
3. Infer a corrected expression matrix.

2 Set up

We will set up libraries necessary for this notebook.

```
suppressPackageStartupMessages({
  library(future)
  library(knitr)
  library(SoupX)
  library(Seurat)
  library(stringr)
  library(tidyverse)
  library(tinytex)
  #library(SingleCellExperiment)
  library(Matrix)

  # Evaluate Seurat R expressions asynchronously when possible using future package
  plan(multisession, workers = parallelly::availableCores())})
```

3 Directories and paths to file Inputs/Outputs

We will set up directories and paths necessary for this notebook.

```
attach(params)
```

```
## The following object is masked _by_ .GlobalEnv:
##
##     root_dir
```

```
analysis_dir <- file.path(root_dir, "analyses", "upstream-analysis")
module_results_dir <- file.path(analysis_dir, paste0("results"))

# Input files
metadata_file <- file.path(metadata_dir, "project_metadata.tsv") # metadata input file
```

[2] https://cran.r-project.org/web/packages/SoupX/vignettes/pbmcTutorial.html
[3] https://doi.org/10.1093/gigascience/giaa151

```
# Create SoupX_results_dir
SoupX_results_dir <- file.path(module_results_dir, paste0("01_SoupX"))
if (!dir.exists(SoupX_results_dir)) {
  dir.create(SoupX_results_dir)}

# Create to plots directory
plots_dir <- file.path(analysis_dir, "plots")

# Create SoupX_plots_dir
SoupX_plots_dir <- file.path(plots_dir, paste0("01_SoupX"))
if (!dir.exists(SoupX_plots_dir)) {
  dir.create(SoupX_plots_dir)}

source(paste0(root_dir, "/figures/scripts/theme_plot.R"))
```

4 Read metadata file and define `sample_name`

```
# Read metadata
project.metadata <- read.csv(metadata_file, sep = '\t', header = TRUE)
sample_name <- unique(as.character(project.metadata$ID))
sample_name <- sort(sample_name, decreasing = FALSE)
print(sample_name)
```

```
## [1] "DYE1472" "DYE2111" "DYE2547" "DYE2548" "DYE2553" "DYE2554" "DYE2557"
```

5 Read in 10x data and run SoupX

We will estimate the contamination fraction `rho`. The posterior distribution is calculated using a Poisson likelihood with a gamma distribution prior, parametrised by its mean priorRho and standard deviation priorRhoStdDev. The dotted line in the above plot shows the prior distribution. The default parameters have been calibrated to be fairly non-specific with a slight preference towards values of rho in the 0% to 10% range which is most commonly seen for fresh (i.e. not nuclear) single cell experiments. The default values place only a very weak constraint, as can be seen by setting a uniform prior.

If contamination fraction estimate is $> 20\%$, it is recommended to set `rho` manually. In this case, we will set 5% background to be removed by default.

```
# Create contamination summary table
contamination_summary <- matrix(ncol = 4, nrow = 1) %>% as.data.frame()
colnames(contamination_summary) <- c("sample_name", "rho_estimated_by_SoupX", "rho_adjusted"

for (i in seq_along(sample_name)) {

  cat("Beginning to process sample", sample_name[i], "\n")

  # Create directory
  samples_SoupX_results_dir <- file.path(SoupX_results_dir, sample_name[i])
  if (!dir.exists(samples_SoupX_results_dir)) {
    dir.create(samples_SoupX_results_dir)}
  print(samples_SoupX_results_dir)
```

```r
#strainedCounts_SoupX_results_dir <- file.path(samples_SoupX_results_dir, "strainedCounts
#if (!dir.exists(strainedCounts_SoupX_results_dir)) {
#  dir.create(strainedCounts_SoupX_results_dir)}
# print(strainedCounts_SoupX_results_dir)

cat("Reading in data\n")

filtered_matrix_file <- file.path(data_dir, sample_name[i], "outs", "filtered_feature_bc_m
raw_matrix_file <- file.path(data_dir, sample_name[i], "outs", "raw_feature_bc_matrix.h5")
clusters_file <- file.path(data_dir, sample_name[i], "outs", "analysis", "clustering", "ge

# Read data
filtered_matrix <- Seurat::Read10X_h5(filtered_matrix_file, use.names = TRUE)
raw_matrix <- Seurat::Read10X_h5(raw_matrix_file, use.names = TRUE)
clusters <- read.csv(clusters_file)

# Save plot
#pdf(file = paste0(SoupX_plots_dir, "/", sample_name[i], "_SoupX_contamination_fraction_p

# Create df
mDat <- data.frame(clusters=clusters$Cluster, row.names=clusters$Barcode)

# Make a "SoupChannel", the object needed to run SoupX
sc_raw <- SoupChannel(raw_matrix, filtered_matrix, mDat)

# Estimate contamination fraction rho
sc <- autoEstCont(sc_raw, forceAccept = TRUE, tfidfMin = 0.1, soupQuantile = 0.9)
print(sc)
soup_fraction_value <- sc$fit$rhoEst

# To not remove more than 20% background
if (soup_fraction_value > 0.2) {
  print(soup_fraction_value)
  cat("Estimated soup fraction >20%, set to 5% by default.\n")
  sc = setContaminationFraction(sc_raw, soup_fraction_value_default)
  contamination_summary[i,] <- c(sample_name[i], soup_fraction_value, "yes", soup_fraction
  print(sc)
  } else {
    cat("Estimated soup fraction\n is ", `soup_fraction_value`, "\n")
    contamination_summary[i,] <- c(sample_name[i], soup_fraction_value, "no", soup_fracti

# Correcting expression profile and clean data
# This will contain a corrected matrix to be used in place of the original table of counts
out <- adjustCounts(sc, roundToInt = TRUE) # roundToInt option to make sure we output int
colnames(out) <- paste0(sample_name[i], ":", colnames(out))
colnames(out) = gsub('-1$','', colnames(out)) # remove -1

#DropletUtils:::write10xCounts(paste(SoupX_results_dir, "./strainedCounts", sep = ""), ou
#write10xCounts(path = file.path(SoupX_results_dir, sample_name[i], "strainedCounts"), x =
#Write10X(out_merge, path = file.path(SoupX_results_dir, "strainedCounts")) #library(Seura
```

```
  #writeMM(out, file = file.path(strainedCounts_SoupX_results_dir, "matrix.mtx"))

  # Write barcodes (column names of the count matrix)
  #write.table(colnames(out), file = file.path(strainedCounts_SoupX_results_dir, "barcodes.
  #           quote = FALSE, row.names = FALSE, col.names = FALSE)

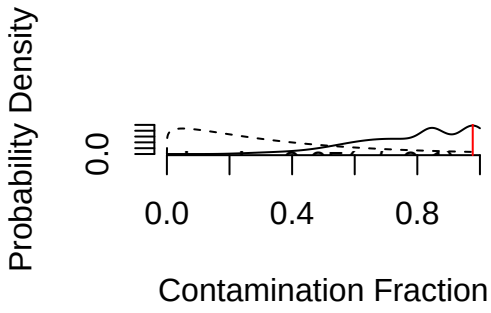
  # Write features (row names of the count matrix)
  #write.table(rownames(out), file = file.path(strainedCounts_SoupX_results_dir, "features.
  #           quote = FALSE, row.names = FALSE, col.names = FALSE)

  saveRDS(out, file.path(samples_SoupX_results_dir, "mtx_corrected_SoupX.rds")) # Save comb

  cat("Processing complete for sample", sample_name[i], "\n")

  # Let's combine corrected matrices from all samples
  # We will use this file in later stage of the `upstream-analysis` module
  #  if(i==1) {
  #    out_merge <- out
  #    } else {
  #      out_merge <- cbind(out_merge, out) }
  #dev.off()
}
```

```
## Beginning to process sample DYE1472
## [1] "/research/dept/dnb/core_operations/Bioinformatics/dyergrp/Chroni_A/Addendum_3D2D_EWS
rna-seq-snap-Addendum-3D2D-EWS-dual-genome/analyses/upstream-analysis/results/01_SoupX/DYE14
## Reading in data
## 17632 genes passed tf-idf cut-off and 7066 soup quantile filter.  Taking the top 100.
## Using 45 independent estimates of rho.
## Estimated global rho of 0.80
```




```
## Extremely high contamination estimated (0.8).  This likely represents a failure in estima
## Channel with 72302 genes and 3307 cells
## [1] 0.8
## Estimated soup fraction >20%, set to 5% by default.
## Channel with 72302 genes and 3307 cells
```

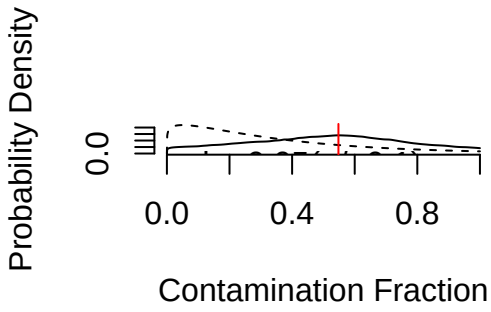
```
## Expanding counts from 3 clusters to 3307 cells.

## Processing complete for sample DYE1472
## Beginning to process sample DYE2111
## [1] "/research/dept/dnb/core_operations/Bioinformatics/dyergrp/Chroni_A/Addendum_3D2D_EWS
rna-seq-snap-Addendum-3D2D-EWS-dual-genome/analyses/upstream-analysis/results/01_SoupX/DYE2
## Reading in data

## 17363 genes passed tf-idf cut-off and 7147 soup quantile filter.  Taking the top 100.

## Using 2669 independent estimates of rho.

## Estimated global rho of 0.55
```




```
## Extremely high contamination estimated (0.55).  This likely represents a failure in estim

## Channel with 72302 genes and 88117 cells

## [1] 0.548
## Estimated soup fraction >20%, set to 5% by default.

## Channel with 72302 genes and 88117 cells

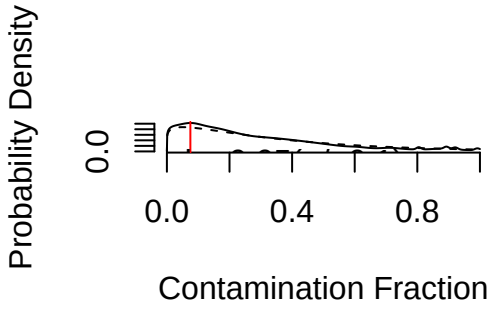
## Expanding counts from 32 clusters to 88117 cells.

## Processing complete for sample DYE2111
## Beginning to process sample DYE2547
## [1] "/research/dept/dnb/core_operations/Bioinformatics/dyergrp/Chroni_A/Addendum_3D2D_EWS
rna-seq-snap-Addendum-3D2D-EWS-dual-genome/analyses/upstream-analysis/results/01_SoupX/DYE25
## Reading in data

## 15296 genes passed tf-idf cut-off and 7163 soup quantile filter.  Taking the top 100.

## Using 273 independent estimates of rho.

## Estimated global rho of 0.07
```




```
## Channel with 72302 genes and 25956 cells

## Estimated soup fraction
##  is  0.075

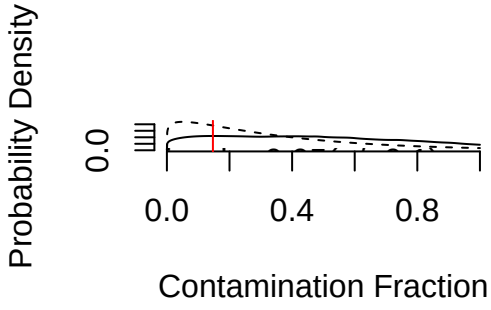
## Expanding counts from 9 clusters to 25956 cells.

## Processing complete for sample DYE2547
## Beginning to process sample DYE2548
## [1] "/research/dept/dnb/core_operations/Bioinformatics/dyergrp/Chroni_A/Addendum_3D2D_EWS
rna-seq-snap-Addendum-3D2D-EWS-dual-genome/analyses/upstream-analysis/results/01_SoupX/DYE25
## Reading in data

## 15886 genes passed tf-idf cut-off and 7004 soup quantile filter.  Taking the top 100.

## Using 1005 independent estimates of rho.

## Estimated global rho of 0.15
```




```
## Channel with 72302 genes and 54192 cells

## Estimated soup fraction
##  is  0.147

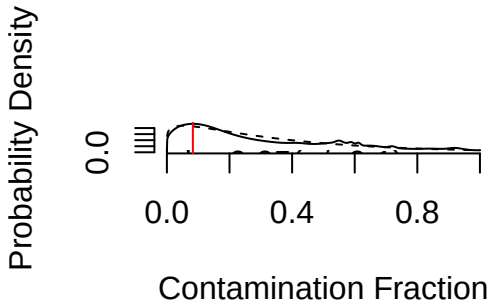
## Expanding counts from 17 clusters to 54192 cells.

## Processing complete for sample DYE2548
## Beginning to process sample DYE2553
## [1] "/research/dept/dnb/core_operations/Bioinformatics/dyergrp/Chroni_A/Addendum_3D2D_EWS
rna-seq-snap-Addendum-3D2D-EWS-dual-genome/analyses/upstream-analysis/results/01_SoupX/DYE25
## Reading in data

## 18640 genes passed tf-idf cut-off and 7122 soup quantile filter.  Taking the top 100.
```

```
## Using 248 independent estimates of rho.

## Estimated global rho of 0.08
```




```
## Channel with 72302 genes and 12430 cells

## Estimated soup fraction
##  is  0.083

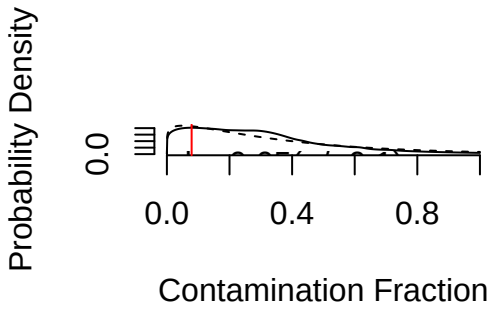
## Expanding counts from 10 clusters to 12430 cells.

## Processing complete for sample DYE2553
## Beginning to process sample DYE2554
## [1] "/research/dept/dnb/core_operations/Bioinformatics/dyergrp/Chroni_A/Addendum_3D2D_EWS
rna-seq-snap-Addendum-3D2D-EWS-dual-genome/analyses/upstream-analysis/results/01_SoupX/DYE25
## Reading in data

## 16554 genes passed tf-idf cut-off and 7179 soup quantile filter.  Taking the top 100.

## Using 893 independent estimates of rho.

## Estimated global rho of 0.08
```



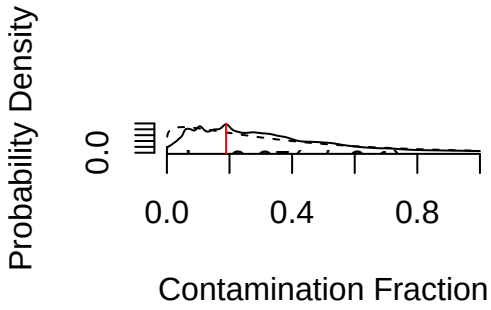

```
## Channel with 72302 genes and 31800 cells

## Estimated soup fraction
##  is  0.079

## Expanding counts from 18 clusters to 31800 cells.

## Processing complete for sample DYE2554
## Beginning to process sample DYE2557
## [1] "/research/dept/dnb/core_operations/Bioinformatics/dyergrp/Chroni_A/Addendum_3D2D_EWS
rna-seq-snap-Addendum-3D2D-EWS-dual-genome/analyses/upstream-analysis/results/01_SoupX/DYE25
```

```
## Reading in data
## 15103 genes passed tf-idf cut-off and 7100 soup quantile filter.  Taking the top 100.
## Using 509 independent estimates of rho.
## Estimated global rho of 0.19
```




```
## Channel with 72302 genes and 25432 cells
## Estimated soup fraction
##  is  0.189
## Expanding counts from 11 clusters to 25432 cells.
## Processing complete for sample DYE2557
```

6 Contamination fraction `rho` per sample

We will summarize the estimated and removed background due to contamination per sample.

Table 1: Summary of contamination fraction per sample

sample_name	rho_estimated_by_SoupX	rho_adjusted	rho_removed
DYE1472	0.8	yes	0.05
DYE2111	0.548	yes	0.05
DYE2547	0.075	no	0.075
DYE2548	0.147	no	0.147
DYE2553	0.083	no	0.083
DYE2554	0.079	no	0.079
DYE2557	0.189	no	0.189

7 Save output files

```
#saveRDS(out_merge, file.path(SoupX_results_dir, "SoupX_corrected_mtx_merged.rds")) # Save

contamination_summary %>%
  write_tsv(file.path(SoupX_results_dir,"SoupX_contamination_summary.tsv")) # Save `contami
```

8 Session Info

```
## R version 4.4.0 (2024-04-24)
## Platform: x86_64-pc-linux-gnu
## Running under: Ubuntu 22.04.4 LTS
##
## Matrix products: default
## BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
## LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.20.so;  LAPACK versi
##
## locale:
## [1] C
##
## time zone: Etc/UTC
## tzcode source: system (glibc)
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] Matrix_1.7-2       tinytex_0.54       lubridate_1.9.4    forcats_1.0.0
##  [5] dplyr_1.1.4        purrr_1.0.2        readr_2.1.5        tidyr_1.3.1
##  [9] tibble_3.2.1       ggplot2_3.5.1      tidyverse_2.0.0    stringr_1.5.1
## [13] SeuratObject_5.0.2 Seurat_4.4.0       SoupX_1.6.2        knitr_1.49
## [17] future_1.34.0      yaml_2.3.10
##
## loaded via a namespace (and not attached):
##   [1] RColorBrewer_1.1-3     jsonlite_1.8.9         magrittr_2.0.3
##   [4] spatstat.utils_3.1-2   farver_2.1.2           rmarkdown_2.29
##   [7] vctrs_0.6.5            ROCR_1.0-11            spatstat.explore_3.3-4
##  [10] htmltools_0.5.8.1      sass_0.4.9             sctransform_0.4.1
##  [13] parallelly_1.42.0      KernSmooth_2.23-26     bslib_0.9.0
##  [16] htmlwidgets_1.6.4      ica_1.0-3              plyr_1.8.9
##  [19] plotly_4.10.4          zoo_1.8-12             cachem_1.1.0
##  [22] igraph_2.1.4           mime_0.12              lifecycle_1.0.4
##  [25] pkgconfig_2.0.3        R6_2.5.1               fastmap_1.2.0
##  [28] fitdistrplus_1.2-2     shiny_1.10.0           digest_0.6.37
##  [31] colorspace_2.1-1       patchwork_1.3.0        tensor_1.5
##  [34] irlba_2.3.5.1          progressr_0.15.1       spatstat.sparse_3.1-0
##  [37] timechange_0.3.0       httr_1.4.7             polyclip_1.10-7
##  [40] abind_1.4-8            compiler_4.4.0         bit64_4.6.0-1
##  [43] withr_3.0.2            MASS_7.3-64            tools_4.4.0
##  [46] lmtest_0.9-40          httpuv_1.6.15          future.apply_1.11.3
##  [49] goftest_1.2-3          glue_1.8.0             nlme_3.1-167
##  [52] promises_1.3.2         grid_4.4.0             Rtsne_0.17
##  [55] cluster_2.1.8          reshape2_1.4.4         generics_0.1.3
##  [58] hdf5r_1.3.12           gtable_0.3.6           spatstat.data_3.1-4
##  [61] tzdb_0.4.0             data.table_1.16.4      hms_1.1.3
##  [64] sp_2.2-0               spatstat.geom_3.3-5    RcppAnnoy_0.0.22
##  [67] ggrepel_0.9.6          RANN_2.6.2             pillar_1.10.1
##  [70] spam_2.11-1            vroom_1.6.5            later_1.4.1
```

```
##  [73] splines_4.4.0          lattice_0.22-6         survival_3.8-3
##  [76] bit_4.5.0.1            deldir_2.0-4           tidyselect_1.2.1
##  [79] miniUI_0.1.1.1         pbapply_1.7-2          gridExtra_2.3
##  [82] scattermore_1.2        xfun_0.50              matrixStats_1.5.0
##  [85] stringi_1.8.4          lazyeval_0.2.2         evaluate_1.0.3
##  [88] codetools_0.2-20       cli_3.6.3              uwot_0.2.2
##  [91] xtable_1.8-4           reticulate_1.40.0      munsell_0.5.1
##  [94] jquerylib_0.1.4        Rcpp_1.0.14            globals_0.16.3
##  [97] spatstat.random_3.3-2  png_0.1-8              spatstat.univar_3.1-1
## [100] parallel_4.4.0         dotCall64_1.2          listenv_0.9.1
## [103] viridisLite_0.4.2      scales_1.3.0           ggridges_0.5.6
## [106] crayon_1.5.3           leiden_0.4.3.1         rlang_1.1.5
## [109] cowplot_1.1.3
```